Chapter 3
Contemporary Trends and Technologies in Research Libraries:
An Overview

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ABSTRACT

This chapter discusses some of the contemporary trends and emerging technologies in research libraries. The chapter begins by explaining the concept of a research library and differentiating between the services offered in traditional versus digital research libraries. The contemporary trends and technologies discussed in this study include big data, blockchain technology, podcast, vodcast, data everywhere, drones, robotics and artificial intelligence (AI) technology, unplugged, makerspace, and the internet of things (IoT). The chapter is important to library users, librarians, and the general public interested in library practices.

INTRODUCTION

A research library is owned by institutions seeking to promote access to scholarly materials and support research. A typical research library contains extensive print and non-printed materials on a broad spectrum of subjects. Therefore, primary and secondary literature can be accessed from a research library. We can refer to libraries in universities and research institutions as research libraries since they contain specialised branches attending to different disciplines. There are two types of research libraries – refer-

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ence and lending libraries. In practice, reference research libraries do not allow external users to borrow/lend from their collections, whereas lending research libraries render lending services to interested users. Research libraries are designed to satisfy the demands of researchers and are, therefore, filled with authentic and high-quality materials. Traditionally, the worth of a research library is determined by its size. Collection size, budget, expenditures, and staff are other input measures representing the library's effort and potential to meet users' demands (Nitecki & Franklin, 1999). These input metrics do not account for how users' requirements and expectations are successfully met.

The impact of research libraries is quantified somehow, especially in terms of users' interaction with materials and services. Users' expectations are set before the interaction, and their impression of the library services provided is assessed. Part of the impact is quantified by querying the gap between expected standards and users' perceptions. Therefore, the user is a critical evaluator of the library's influence. Through digitisation programmes and agreements for licences that allow subscription collections, research libraries can make computational use of collections easier. As the knowledge of the public regarding data visualisation and analytics develops, so does the number of new users (including librarians) who wish to experiment with computational tools. Many of these users are only at the beginning of their learning curve. Internally, research libraries have many opportunities to apply data-centric techniques like machine learning to augment descriptive information and produce text corpora (Calvert, 2020). This chapter discusses the library services offered in the digital age and the contemporary trends and technologies in research libraries. Following completion of this chapter, the reader should be able to:

- Understand and explain the meaning of a research library.
- Differentiate between library services offered traditionally and digitally.
- Identify and comprehend trends and technologies used in research libraries.

LIBRARY SERVICES IN THE DIGITAL AGE

We live in the digital age, and in many situations, the central role of information in this era is digital (Mayega, 2008). There is increased Internet usage for information spread and retrieval in today's world, which has affected the way knowledge is gathered and used (Owan et al., 2021; Owan & Asuquo, 2021). The shift toward electronic research libraries has dramatically altered the public's views towards considering conventional libraries. The digital network is the principal mechanism for sharing information. Information across different file formats (such as audio, text, video and audio-visuals) is now easily created, managed, stored and accessed in ways that were only imaginable some decades back. Emerging technologies enable library professionals to identify, gather, organise, customise, and distribute information products and services to the user community in various forms and kinds, both on-demand and in advance, in physical and virtual locations, in real-time (Ayo-Olafare, 2020). Consequently, many libraries are spending to buy digital library materials to satisfy modern-day expectations.

There are several ways in which a research library can utilise the Internet, ranging from setting up a library website to allowing consumers to browse existing catalogues electronically. Scholarly materials such as published research papers may be found using specialised search engines. People may do research on the Internet using the computers and Internet connectivity available in public libraries (Mostapha, 2005). Users under the age of 30 are especially drawn to the convenience and accessibility of online library resources (Corradini, 2006; Snowball, 2008). Most libraries are significant partners for google search engines because of their extensive collection of relevant information and have gained corresponding advantages in circumstances in which they have brokered (Darnton, 2009). Digital research libraries have shifted their focus from primarily offering print materials to delivering more internet connectivity. A variety of issues confront research libraries in adjusting to new methods of service delivery (Abram & Luther, 2004). This makes information literacy capabilities less critical. The fall in the use of libraries, especially in reference services, raises doubts about the need for these services. For libraries to thrive with the Internet and avoid losing customers, library researchers have admitted that they must improve their advertising campaigns. In addition, this involves advocating the importance of information training opportunities in the library community.

DIGITAL SKILLS NEEDED IN THE DIGITAL TRANSFORMATION ERA

Many branches of study, including the humanities, have been touched by digital transformation. Digital transformation is a trend that corporations have been aware of for several years and as technology advances at an astounding rate, digital transformation moves businesses forward. It occurs when businesses recognise the value of social learning in the design and delivery of information, problem-solving, knowledge sharing, and user-generated content (Sousa & Rocha, 2018). However, in the twenty-first century, the variety of digital transformation applications is astounding. Speed is no longer the only advantage, as digital technology's ease, power, variety and speed affect everything from interactivity to accessibility (Fonseca & Picoto, 2020). The rise of new digital technologies is compelling businesses to transform their products, services, workflows and business models (Moroz, 2018). Enterprises were the first to recognise the primary benefit of digital technology speed when computers initially made their way into offices. Nowadays, the increasing number of research libraries are also positioning themselves to take advantage of these advancements and acquire a competitive advantage.

However, the number of potential digital service applications and software types and their interaction with users is so vast that experts with the appropriate digital skills are required to use all emerging digital tools in this new digital era. Our lives are more reliant on technology, and as our reliance on the Internet and digital communications grow, our workforce must adapt to meet the changing skill demands. For decades, digital skills have been required in the workplace. Digital skills are defined as using digital devices such as computers and smartphones to search, assess, use, share, and create content. There has always been a demand for digitally inclined individuals, such as IT specialists and staff who can securely eject a floppy disc, as long as computers, servers and electronic communications have existed. All organizations in the digital economy, including academic and research libraries are coming to realise that digital skills are vital for their employees in the digital transformation era. Research libraries want the great majority of their library staff and team, not just a select few, to have solid digital abilities to stay relevant in this digital era. The library staff and team must have the following digital abilities for a successful digital transformation.

- **Data Analytics:** Data and analytics are the primary accelerators of an organisation's digitisation and transformation activities. Any digital transformation effort will generate a large amount of data. As a result, the most in-demand skill sets should be data analytics, market research analytics, and database administration (Saha, 2018).

Figure 1. Digital skills for digital transformation

- **Digital security:** Digital security is the umbrella word for the resources used to safeguard your online identity, data, and other assets. Web services, antivirus software, smartphone SIM cards, biometrics, and encrypted personal gadgets are examples of these tools.
- **Digital marketing:** Digitally aware marketers who understand and capitalise on the degree of involvement provided by digital marketing have the best chance of creating an impression on consumers. Social media, video creation, email marketing, and content marketing are all effective channels of promotion that necessitate a well-rounded skill set.
- **Cloud Computing:** Cloud computing is an excellent tool at their disposal to help them expedite their transformation efforts. Cloud computing empowers businesses to collaborate with cloud service providers and provides intelligent, readily available, adaptive, and flexible digital solutions.
- **Critical thinking:** The ability to assess material logically and make a reasoned decision based on your analysis is known as critical thinking. Critical thinkers refuse to take things at face value and are conscious of their own cognitive biases, allowing them to arrive at objective judgments.
- **Creativity:** Creativity is the capacity to approach a task or problem in a new or unique way or use one's imagination to develop novel ideas. Creativity allows you to solve complex problems or create novel approaches to assignments.
- **Big Data:** The significance of big data in digital transformation stems from an organisation's ability to combine both in its attempts to enable both digitalisation and automation of business activities. This digitalisation and automation drive efficiency, innovation, and the development of new business models.
- **Social media:** Social media skills assist professionals in developing and implementing marketing initiatives to promote a company. To be successful, you must have a unique flair and an awareness of what makes material shareable, and your social media skills description should reflect these attributes.
- **Machine Learning:** Machine learning requires computation on enormous data sets. Therefore, solid foundational abilities in computer architecture, algorithms, data structures, and complexity are required. Getting into the programming books in depth and trying new things will be beneficial.

- **Blockchain:** The Blockchain Platform will be critical in ushering businesses into the digital transformation age. By providing a means to secure data exchange, dynamic apps, and decentralised identification, the Blockchain Platform enables digital transformation use cases. It allows businesses to improve application modernisation by automating business operations, launching new digital products, and monetising essential digital content (Prakash & Ambekar, 2020).

CONTEMPORARY TRENDS AND TECHNOLOGIES IN RESEARCH LIBRARIES

Numerous trends and technologies are implemented by various research libraries worldwide. The prominent trends and technologies in research libraries include Big Data, Blockchain, Podcast, Vodcast, Data Everywhere, Drones, Robotics, Artificial Intelligence Technology, Unplugged, Makerspace, and the Internet of Things. The world is continually evolving, and new technologies are adopted regularly (Shashikumara et al., 2019). Library professionals must identify and comprehend future library trends and technologies to plan and implement new solutions (Axiell Group AB, 2016). Trends and innovations that impact society and research libraries are no exception. This chapter discusses the most important trends and technologies in research libraries below.

Big Data

Big data is large-scale data that continue to grow without decreasing, taking different forms. Every second, a massive quantity of data sets is generated from all over the globe, meaning that the volume of data will never decrease but will instead continue to grow. To acquire, curate, manage, and analyse such datasets reasonably, commonly used software technologies cannot keep up with the growing amount of big data. Based on the developments in research on big data, it has been documented that big data is characterised by 17 V's and one C (Arockia et al., 2017). These attributes include voluntariness, volume, volatility, vocabulary, visualisation, viscosity, virality, versatility, verbosity, veracity, venue, velocity, variety, variability, value, validity, vagueness, and complexity.

Large volumes of data may be collected, stored, manipulated and analysed by research libraries using big data technologies. Books, e-books, online journals, research papers, conference proceedings, seminar papers, institutional repositories, and other materials make up most of the library's collection. Electronic and physical setups are specifically designed for research scholars and clients to meet their data requirements. Large-scale database analysis is required to keep up with the never-ending inflow of data and knowledge. Libraries are great for benefiting from Big Data while also delivering analytic library services to users. Researchers may now use library services such as data management strategy, data collection, data curating, and information archival (Sugimoto et al., 2012).

Today, "Big Data" is mentioned commonly in news headlines and other published content (Wittmann & Reinhalter, 2014). Big data is gaining traction in the data and correspondence period (Ball, 2019). Currently, "Big Data" is a developing topic that poses several IT issues regarding data gathering, storage, search, structure, and visualisation (Garoufallou & Gaitanou, 2021). While we as librarians have a few practical and powerful strategies for managing this type of data, there has not been much research on using metadata to create computerised resources using apparatuses such as big data and distributed computing innovation (Panda, 2021). For research libraries, Big Data analytics faces two main challenges: first, the sheer nature of the data needs an enormous amount of computational power and storage; second, the analytical methods and algorithms themselves are complex, making Big Data analytics a computationally demanding endeavour (Al-Barashdi & Al-Karousi, 2019). Cloud has proven to be the most suitable infrastructure solution for supporting Big Data analytics applications' storage and processing needs.

Blockchain Technology

The blockchain is a distributed, tamper-resistant data storage system (Singh et al., 2022). This is in line with librarians' usual practices of gathering, preserving, and disseminating authoritative information. Blockchain is a relatively new and dependable technology that aids in data protection, preservation, and reliability (Abid, 2021). Libraries are 21st-century change agents, integrating new technologies to deliver maximum information in the shortest time. Blockchain technology may be utilised to address several issues in the library sector, such as securing information in a distributed, tamper-resistant framework. The blockchain ledger may be used for various applications outside financial markets, including tamper-proof publications, distributed ownership records, and academic archives in libraries (Hoy, 2017).

Blockchain technology can also enable the smooth borrowing of research materials among libraries and between libraries and users. Issues of borrowing without proper record-keeping will be forgotten stories with the implementation of blockchain technology in research libraries. For instance, the COVID-19 experience coupled with other social changes occasioned homelessness, statelessness, employment and travel, resulting in groups of individuals who are permanently or temporarily displaced from their homes. Research libraries in the digital era must adapt to accommodate them. Refugees and migrants of all ages need access to information resources that may assist them in developing new skills and finding jobs. Libraries may provide services that these communities cannot get elsewhere, yet they often are unable to obtain a library card or borrow goods. Developing an interoperable blockchain-based system that connects all library systems and a secure, verifiable digital identity that can be used to access information at participating libraries.

The blockchain will allow unlimited access to digital content and print collections for all participants to secure users' privacy and personal identity. The International Federation of Library Associations (IFLA) might prototype a global interlibrary loan system with blockchain technology. Due to the nature of interlibrary loans and the foreign currency transactions involved. Blockchain technology may evaluate the validity (correctness) of data over time. Some scholars successfully showed the feasibility of a low-cost, independently verifiable technique for auditing and certifying the legitimacy of scientific investigations (Irving & Holden, 2016). They accomplished this by generating a new private Bitcoin key from the plaintext of a protocol trial paper. As a result, researchers can instantly verify the blockchain's timestamp. When comparing the original record's hash to the blockchain's, the records have been altered if the two hashes do not match.

Podcast

A podcast is an entertaining method to advertise library collections services to attract new readers (Lee, 2006). Podcasting provides libraries with intriguing opportunities to communicate with their users in a new and well-known way. It can help remove barriers, relieve pressure or anxiety students may have about using the library and put a welcoming face on the library. Libraries can take advantage of the popularity of podcasting and use it to spread information to a larger audience. Clients have another option to receive ready-made knowledge in a compact format using podcasting technology in library services, which develops a mobile learning environment. Due to the importance of many libraries, such as the Los Angeles public library, London library, Library of Congress, the British Library, DC library, free library Philadelphia, Kiama Library, and the New York library, among others, are now using podcasting service to organise and promote otherwise difficult-to-find scholarly and recreational items. It is expected that librarians will walk away with direct marketing and organising ideas and ideas for how to promote and organise this relatively new medium in the years ahead. Libraries can also use podcasts to host authors and well-known citizens for various annual activities. In order to listen to these activities, individuals may either download one of the audio programmes listed or subscribe to the libraries' authors and events podcasts.

Vodcast

Vodcast is derived from the root term "Vod", which is an acronym for "video on demand" (Kolbitsch & Maurer, 2006). Vodcasts, also known as video podcasts, are an extension of the development of podcasts, albeit they differ in that they are not just audio in nature but also visual (Naidu & Constable, 2016). Vodcasts are, by definition, a collection of audio-visual materials that may be downloaded to a personal computer or other video-enabled connected devices through the Internet (Greef, 2012). Vodcasts need more storage space due to the audio and video components. Libraries utilise vodcasts to show their accomplishments and attract the public to check out their upcoming events. Videos of library programmes and activities have become a popular way for libraries to demonstrate their successes while also encouraging attendees to future events. Teachers and students can connect effectively outside a classroom in real-time.

Along with increasing lecturer-student interaction, the vodcast reduces the need for face-to-face counselling by extending the relationship beyond classroom time (Sutton-Brady, 2009). The use of vodcast is an effective way of reducing face-to-face interactions, especially now that social distancing is encouraged due to the Covid-19 pandemic. Due to its importance, many scholars are now interested in how students' learning outcomes can be enhanced through vodcasting (Backhaus et al., 2019; Goundar & Kumar, 2021; Yeganeh & Izadpanah, 2021).

Data Everywhere

Data Everywhere was founded by Dan McGee and Dave Bortz in November 2013 and is in Chicago, Illinois. Data Everywhere is an add-in for spreadsheets that enables sharing specific datasets with other spreadsheets, databases, websites, and mobile devices. It connects existing spreadsheets to a cloud-based data management service that provides replication, security, revision control, basic workflow, and search for these applications. Ordinary business users may share their data with co-workers and other parties as raw data rather than as files that have been embalmed, enabling other users to remix the data in novel ways. As opposed to a mere document, Excel as a data source enables Data Everywhere users to create new sorts of applications that would have previously required specialised programming. Data collection and management are critical in today's libraries. New technologies provide excellent opportunities to collect, store, and analyse accurate user data and personal information. Data can be accessed using mobile phones, iPads, and other internet-connected devices. Data can be used to produce products and services, boost marketing, and promote information in the news. Data services have evolved into essential information services for libraries, allowing researchers to link with research from other studies (Library of the Future, 2018).

Organisations of all sizes, from large corporations to start-ups, will need to be adept at capturing, storing and consuming data. This must be a never-ending data cycle in a feedback loop. This data cycle can be addressed with a slew of tools and technologies, but their effectiveness is contingent on the direction and governance provided by the organisation. Work on your data strategy always, as this will help your company improve its overall performance, empower employees, and provide more control over their data and data-driven decisions. There are several tools available for businesses to use after having enough data. Organisations may encounter several obstacles when implementing and managing their data strategy (e.g., issues might arise from architecture, data infrastructure, competencies, operational expenses, business user expectations, people, etc.) as they go down the data analytics journey lane. These issues must be addressed and resolved as soon as possible.

Drones

Drones, also known as unmanned aerial vehicles (UAVs), are becoming more popular in various industries, including leisure, security, military, agriculture, energy, insurance, and hydrology. Robots that fly in the air, called drones, perform many tasks such as taking pictures, recording films, and gathering data in various formats from their surroundings. Fixed-wing and multirotor drones are the two main categories. Multirotor drones can operate in places where people cannot, gather data in several formats, and even intervene when necessary due to their portability and size (Natarajan et al., 2018). In addition, multirotor drones with protective hulls are very durable in collisions, making them even more helpful for exploring unexplored territory. Parcel delivery services and other organisations use flying robots (Gatteschi et al., 2015). Though several well-known drone variants have been in use for over two decades, the public's attention has been drawn to newer models and technologies that have not been around as long. Applications of drones have experienced a significant paradigm shift. Even while drones have become an essential part of contemporary warfare, their use for non-military purposes has increased significantly. Consequently, drones are now used in other areas such as aerial photography and filming, shipping, search and rescue, geographic mapping, wildlife monitoring, law enforcement, crop inspections, pollution inspections, and meteorological services.

Drones may be used at the library because of their wide range of non-military applications. It's difficult for folks in today's fast-paced world to visit the library at their convenience. The fact that you are in the wrong place at a bad moment further compounds the problem. People who live near a library or have access to a time-saving route away from regular heavy traffic may use these advantages, but this is not the case for everyone. Drones may be used for a variety of purposes. Using drones to deliver books and other library resources to users might be a game-changing feature of this new technology (Nath, 2018). The first and most important prerequisite for a library to provide a drone delivery service is a library mobile app or a mobile-friendly library website (Saloi, 2021). Drones are helpful for library organisations (Nath, 2018). Modern flying robots outfitted with technology can transport books, furniture, and other items from one location to another. Using the app or website, library patrons may borrow items from the collection; however, this needs a user login and the ability to reveal their current location and delivery address. It may also be used as a security measure both inside and outside the library and is significantly superior to fixed cameras because it moves and covers the entire area, providing 24-hour observation. The preponderance of national and district libraries is in metropolitan regions. Many people no longer find it easy to visit the library as cities develop, despite its central location, which was meant to promote exposure and accessibility. Population expansion in the city has increased the number of library customers who travel far to visit. Persons who can only attend the library after work, people who can't visit for personal or professional reasons, traffic congestion, or the absence of public transit or direct routes to the library may need many taxis or buses to reach the library (Nath, 2018). This limitation may be overcome by sending books to a specified location, generating new potential for libraries and users. The procedure begins with a patron requesting a book through a specific library app, which is then received and processed by the library's concerned section. The ordered book is carried to the drone department, where a skilled pilot attaches the book to the drone and allows it to fly to the patron's hand. It automatically returns to the origin after delivery, with all essential operations and advancements updated for both sender and receiver.

Robotics and Artificial Intelligence (AI) Technology

Libraries are progressively adopting robotics and AI technology (Harada, 2017). AI and robotics are powerful for automating activities within and outside industrial facilities. In recent years, AI has become more prevalent in robotic solutions, bringing adaptability and learning capabilities to previously inflexible applications (Vysakh & Rajendra, 2019). This highlights how libraries can combine robotics with other AI technologies, such as using a drone controlled by a robot to keep the library under constant observation. As a user help and guide, talking robots can be put throughout the library.

The application of AI in libraries is to construct computers or robots that can think, act and, in some situations, surpass human intellect. This has been shown to have obvious implications for librarianship (Omame & Alex-Nmecha, 2020). Consequently, many public and research libraries are now using artificial intelligence for different purposes. Some options include expert systems for reference services, robots that can read books and shelves, and virtual reality for immersive learning. Cataloguing and topic indexing, reference services, technical support, and shelf-reading are all instances of artificial intelligence in library systems.

Unplugged

Unplugging can mean various things, but it always implies a break from the fast-paced, overstimulating, and sometimes frenetic world of technology that propels so many of us through our days. Unplugging, then, implies taking a break from a typical routine and creating a quieter zone to engage in undistracted action, engagement, or inaction. Libraries will continue to be significant sources of technological access. At the same time, libraries are all about making connections (ALA Libraries Transform, 2018). People often need to unplug to create those connections, and libraries have developed creative ways to assist users. Unplugged gives students access to high-quality reference materials to help them study. This brings the library into the classroom and the house, always making it available in e-books on any device.

Makerspace

The Makerspace, a combination of two notions ('Maker' and 'Space'), is an offspring of the technology era's innovativeness, symbolising a blending of two worlds (Omosebi & Bakare, 2022). Makerspaces, also known as Hackerspaces or FabLabs, are self-contained spaces where people may create, develop, and study. 3D printers, software, electronics, supplies and hardware, craft equipment, and other goods are regularly found in libraries (Public Library News, 2012). Makerspaces are becoming more frequent in public, academic, and research libraries as a creative method to engage users and provide value to traditional library services.

Internet of Things (IoT)

IoT is a new technical change that library and library personnel should be familiar with since it will help libraries improve their facilities, tools, and experience. Libraries will be able to add more value to their services and deliver a better library experience to their clients. Because items are individually identifiable, the Internet of Things is about digitally connecting them. Radio Frequency Identity (RFID), which does the same thing by communicating with computers, marks and updates library management systems with entries to books given to a customer, is still familiar to librarians. However, the difference with IoT is that the Internet interacts with a thing or entity, such as a book. The Internet of Things (IoT) can assist address some of the most common library concerns, such as object misplacement and use, by connecting books, articles, CDs/DVDs, theses, and other physical items (Singh et al., 2022).

The IoT may be used in libraries to provide patrons with easy access to both traditional and online collections and data and directories. For example, librarians and library users may find it easier to discover actual artefacts and explore virtual resources with this technology. Contextual recommendations and information about resources relevant to the current user's interests might likewise be sent using this method. Other aspects of library service, such as consultation and training, might benefit from the IoT. Users' moods, daily habits, and other similar details gleaned from their mobile devices through the IoT might be utilised to create personalised training programmes (Qin, 2018). The IoT may also alert users to the lack of space in the reading room or the absence of workstations. Other commercial IoT applications exhibited more exciting prospects when compared to various library activities. Although the IoT has the potential to be utilised often in marketing and promotion, its value does not seem to be limited to this area. Other uses include process optimisation, library workflow structuring, and the development of unique economic models that make libraries more fun for users and other stakeholders. For a library to be seen as a contemporary institution, it is essential to use IoT-based marketing tactics to promote its operations. Internal libraries might benefit from this technique as well. For example, resource collection, description, analysis, intelligent construction, and suitable resource storage are functional areas of IoT in research libraries.

FACTORS MILITATING AGAINST THE ADOPTION OF THESE TECHNOLOGIES

Almost every organisation faces difficulties with technology adoption. Indeed, presenting new technology to an organisation might be more complicated than the technical installation for most professional services firms (Kieth, 2019). Although technology has made our lives easier, the ease with which we may access shared information has legal implications for businesses. The following are some of the obstacles to new technology adoption:

- **Adopting new technology solely for novelty:** While many new technologies are beneficial, some of it is purely commercial. Even if the technology is helpful, it may only be valuable for specific businesses. Before purchasing any new technology, properly research it and make sure you can use it. Otherwise, you risk having yet another costly paperweight or squandering disc space on applications that you never use.

- **New Technology's disruptive impact:** New technology is disruptive in the near term. Only by integrating with legacy enterprise systems can it reach its full potential. Existing policy and processes will need to be upgraded, causing substantial disruptions. When exposed to real-time situations, the new implementation may generate bugs. It takes some time for it to settle.
- **Resource scarcity:** Emerging trends and technologies are not inexpensive. Even with open-source software, the initial development expense might be costly. The costs of training, inconvenience, and downtime are all indirect costs. During the challenging changeover period, customer dissatisfaction is also dangerous. Such costs overwhelm businesses, causing them to delay technology adoption.
- **Skill challenges:** Emerging technologies, such as Artificial Intelligence, face significant skill shortages. New technology adoption is further hampered by a lack of thought leadership and risk-taking at the highest levels.
- **High cost:** Adopting new technology has numerous advantages. Many businesses want to embrace digital transformation, but they do not believe they have the resources to purchase and integrate new technologies. Successful technology adoption necessitates some initial expenditure for onboarding and deployment.
- **Integration and training:** It takes time and money to train users on new technology and software. Lack of adequate and specialised training on the technology to be adopted is another major factor slowing down the process (Calibraint, 2021). Employees must be trained even if a new technology purports to be "user-friendly". Because not everyone can simply adapt to new technologies, assistance is required to ensure everyone is on board (Scalzo, 2019). It also aids in the prevention of costly mistakes made by first-time users.
- **Onboarding:** Many people mistake focusing solely on the initial implementation of new software or technology, but their training needs do not end there (Altadonna, 2021). You will need to develop a good strategy for introducing new consumers to your technology.
- **Time:** Adapting to new technologies is a lengthy and often demanding process. It can be disappointing not to see the results of your efforts right away.

CONCLUSION

Libraries are an essential aspect of the society in which they exist. Libraries have boosted their services by embracing various technological tools or aids because of the enormous advances in ICT. This period has seen massive information communication and technology changes, and many fields are attempting to adapt to this new environment by implementing various technologies (Saloi, 2021). Libraries are not far behind, either, as they are constantly contending with the changing environment. If scholars are increasingly familiar with the tools offered to them and their involvement is more formalised, research libraries might play a more significant role in the future. Research libraries can utilise their expertise to assist researchers in improving the quality of their grant applications and increasing the institution's success in attracting research funding. Big Data is also one of the most rapidly developing technologies today. Research has documented that the application of Big Data in libraries can assist in information generation, and productivity is higher in libraries that have implemented big data than those that have not. To summarise, Big Data tools and analysis tools necessitate specialised talents and enable more outstanding performance in today's competitive market (Kamupunga & Chunting, 2019). Academic and research libraries are anticipated to become more attractive, relevant, and acceptable because of these new offerings and ongoing improvements. However, libraries' approaches, applications and concepts will continue to evolve.

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KEY TERMS AND DEFINITIONS

Artificial Intelligence: AI is the capacity of a system or a robot computer-controlled to do activities usually performed by humans.

Big Data: A big data set is one that is too large or complicated for standard data processing techniques.

Blockchain: A blockchain is an electronic database shared by computer network nodes.

Digital Age: This is when advances in computer technology have made enormous amounts of data freely available to practically anybody.

Internet of Things: This is a network of physical things connecting and exchanging data over the Internet or other communication networks.

Library Services: These are services provided to increase the use of library items, connect library users with library resources, and meet their information needs.

Research Library: A research library contains diverse scholarly materials produced from research and for the furtherance of other studies in each discipline.

Trends: Trends refer to new ways in which an event, occurrence, phenomena, construct, or technology evolves in newer ways than was previously conceived.